

TAB 4



Board of Education Report

File #: Rep-165-23/24, **Version:** 1

Resolution to Make Certain Determinations and Findings pursuant to Government Code Section 4217.10 Et Seq. and for Authorization to Enter into Energy Service Contracts with Ameresco, Inc and Centrica Business Solutions Services, Inc. for Solar Photovoltaic System Projects at 14 School Sites (PUBLIC HEARING)

January 23, 2024

Procurement Services Division, Facilities Services Division and Office of Eco-Sustainability

Action Proposed:

Adopt a Resolution and authorize the Chief Procurement Officer and/or his designee(s) to execute Agreements, and/or any other reasonable instruments, with Ameresco, Inc. and Centrica Business Solutions Services, Inc. (Centrica) for Solar Photovoltaic (PV) System Projects (Projects) at 14 school sites, as described in Exhibit A.

The Projects will be funded by Ameresco, Inc. and Centrica (Energy Contractors). The Energy Contractors will design, secure agency approval, install, commission, maintain and guarantee production of all-inclusive, fully operational, Division of the State Architect (DSA) approved solar PV canopies/carport and roof-mounted systems at 14 school sites. In addition, Energy Contractors will install infrastructure for future electric vehicle (EV) charging stations at nine sites and battery energy storage systems at five sites.

The Los Angeles Unified School District (District or Los Angeles Unified) will begin making payments to the Energy Contractors after permission to operate the PV systems has been secured from the District's utility providers, Los Angeles Department of Water and Power (LADWP) and Southern California Edison (SCE). The total value of the Agreements is approximately \$98.4 million to be funded by the General Fund over 25 years for the 14 school sites.

The total estimated 25-year energy costs for the 14 school sites is \$133.5 million; the expected net energy cost savings from on-site electricity generation is approximately \$35.1 million over a 25-year period, as set forth in the Resolution (Exhibit B). These Agreements are being procured pursuant to the energy savings/services provisions in Government Code section 4217.10 et seq., as further indicated in the Resolution and will help to achieve Los Angeles Unified's clean energy goals and advance the Los Angeles Unified Strategic Plan.

Background:

In April 2022, Los Angeles Unified issued a Solar Photovoltaic (PV) Systems - Phase1 Request for Proposals (RFP) in support of efforts related to the Transitioning Los Angeles Unified School District to 100% Clean, Renewable Energy Resulting in Healthier Students and More Sustainable, Equitable Communities Board Resolution. The Board Resolution commits Los Angeles Unified to the goals of achieving 100% clean, renewable energy in its electricity sector by 2030 and in all energy sectors, including heating, ventilation, air conditioning (HVAC), cooking, and transportation, by 2040.

The installation of PV systems to generate electricity from solar power offers the District the opportunity to significantly offset its energy costs over the long term and to achieve its environmental stewardship and clean energy goals. Under the RFP, Los Angeles Unified sought proposals from firms to design, secure agency

approval, install, commission, maintain and guarantee production of all-inclusive, fully operational, DSA-approved PV canopies/ carport and/or roof-mounted systems at 14 school sites.

The school sites were selected within the geographic areas of two different utility providers, Los Angeles Department of Water and Power and Southern California Edison, to explore the different types of delivery and financing methods that could offer the greatest benefit for the PV system installations. Other factors considered were sites that had been identified for future roof replacement, and sites with newly installed roofs to avoid roof warranty issues and minimize roof penetrations that could cause water leakage.

Staff anticipates that construction activities related to the installation of the PV systems for all sites will begin in the fourth quarter of 2025 and that all sites be completed by the fourth quarter of 2028. Following completion, the Energy Contractors will secure permission to operate, which is the local electric provider's authorization to interconnect the PV system to the electrical grid.

Expected Outcomes:

Approve the proposed action, which will permit activities in support of Solar PV System Projects to be undertaken at 14 school sites. Staff estimates that the Projects will optimize the District's energy generation, conservation, and cost savings pursuant to California Government Code §4217.10 et seq. A cumulative utility cost savings of approximately \$35.1 million is anticipated over the life span of the equipment due to on-site electricity generation from the PV systems and benefits of the battery energy storage systems at five sites and electric vehicle charging stations infrastructure at nine sites. The anticipated savings in energy costs is set forth in Exhibit B.

Board Options and Consequences:

Adoption of staff's proposal will permit the District to proceed with the implementation of Solar PV System Projects at 14 school sites as part of the pathway for achieving the District's goal of 100% Clean, Renewable Energy Resulting in Healthier Students and More Sustainable, Equitable Communities Board Resolution.

If the proposed action is not adopted, the Projects will not be implemented, and the District will be unable to realize the anticipated cost savings of approximately \$35.1 million. Furthermore, staff would not be able to evaluate the Projects to determine applicability at other school sites in order to reach the District's 100% clean, renewable energy goals in its electricity sector by 2030.

Policy Implications:

The requested action does not change current District policies. Staff's proposal is consistent with Los Angeles Unified's clean energy goals and advances the 2022-2026 Strategic Plan Pillar 4 Operational Effectiveness Modernizing Infrastructure and supports the Transitioning Los Angeles Unified School District to 100% Clean, Renewable Energy Resulting in Healthier Students and More Sustainable, Equitable Communities Board Resolution.

Budget Impact:

Project costs associated with the Solar PV System Projects will be funded by Energy Contractors. The aggregate value of the Agreements is approximately \$98.4 million (\$31.4 million for Ameresco, Inc. and \$67 million for Centrica). After Energy Contractors have secured permission from the utility providers to operate the PV systems, the District will make payments to the firms based on the funded construction costs and annual maintenance and Performance Guarantee amounts. The payments, estimated at \$98.4 million over 25 years for the 14 school sites, will be funded by General Funds. This action is projected to provide approximately \$35.1 million in net cost savings to the District in utility bills over the 25-year period, as set forth in Exhibit B.

Student Impact:

Once completed, the Solar PV System Projects will help reduce energy consumption, thus lowering the utilities cost impact to the General Fund and allow for more financial resources to be allocated to support educational programs. Additionally, clean renewable energy technologies at schools, such as carport and rooftop solar PV systems, battery energy storage systems and Level 2 EV charging stations, provide an educational opportunity for students to learn about real-world energy issues. Such project-based and hands-on experience can help further their STEM education.

Equity Impact:

The school sites were selected within the geographic areas of two different utility providers, Los Angeles Department of Water and Power and Southern California Edison, to explore the different types of delivery and financing methods that could offer the greatest benefit for the PV system installations.

Issues and Analysis:

The proposed action will enable Los Angeles Unified to install up to 10.6 megawatts (MW) of solar power on 14 school sites within the framework of a 25-year turnkey agreement with operations and maintenance included in the fee for services. (A turnkey agreement is a contract for design, construction, installation, permitting, operations, and maintenance, whereby the contracted entity assumes total responsibility for the project.) The Solar PV Projects require compliance with all laws, rules, regulations, and standards for public school construction, the cost of which will be borne by Energy Contractors, Ameresco, Inc. and Centrica.

The overall projected net cost savings to the District is approximately \$35.1 million over 25 years, as provided in the Resolution. At the conclusion of a 25-year Agreement with Ameresco, Inc. and 20-year agreement with Centrica, the District will have ownership rights to the Projects installed at all 14 sites. The District will also have the option to purchase the Projects prior to the end of each agreement without prepayment penalties.

Government Code section 4217.12 allows public agencies to use the procurement method it determines is in the best interests of the public agency. To satisfy the requirements of Government Code section 4217.12, the Board shall adopt the proposed Resolution, making findings and determinations of the energy cost savings, at a regularly scheduled public hearing of which public notice is given at least two weeks in advance. Public Notice of the proposed Resolution and Public Hearing was provided at the December 12, 2023, Meeting of the Board of Education.

Attachments:

Exhibit A - List of School Sites

Exhibit B - Resolution to Make Findings Pursuant To Government Code Section 4217.10 Et Seq., and
Authorize Entering into Energy Services Contracts with Ameresco, Inc. and Centrica Business
Solutions Services, Inc.

Informatives:

Not Applicable.

Submitted:

12/11/23

RESPECTFULLY SUBMITTED,


ALBERTO M. CARVALHO
Superintendent

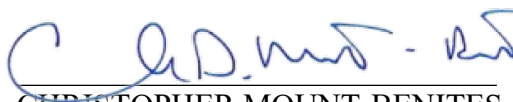
APPROVED BY:


PEDRO SALCIDO
Deputy Superintendent,
Business Services and Operations

REVIEWED BY:


DEVORA NAVERA REED
General Counsel

☒ Approved as to form.

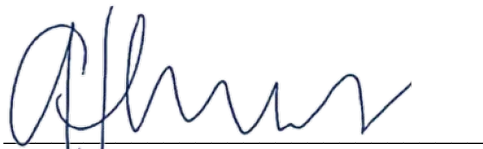
APPROVED & PRESENTED BY:


CHRISTOPHER MOUNT-BENITES
Chief Procurement Officer
Procurement Services Division

REVIEWED BY:


NOLBERTO DELGADILLO
Deputy Chief Business Officer, Finance

☒ Approved as to budget impact statement.

APPROVED AND PRESENTED BY:


KRISZTINA TOKES
Chief Facilities Executive
Facilities Services Division

APPROVED AND PRESENTED BY:


CHRISTOS CHRYSLIOU
Chief Eco-Sustainability Officer
Office of the Superintendent

Exhibit A
List of School Sites

BD	REG	Energy Contractor	School Site	Utility Provider	PV System Size (MW)	EV Charging Infrastructure	Battery Storage
1	South	Centrica	Crenshaw Magnets: Science Tech Engineer Math & Medicine	LADWP	1.207	Yes	No
1	South	Centrica	John Muir Middle School	LADWP	0.612	No	No
1	South	Centrica	Susan Miller Dorsey Senior High	LADWP	0.846	No	No
2	East	Centrica	Benjamin Franklin High School	LADWP	0.763	No	No
3	North	Ameresco, Inc.	Northridge Middle School	LADWP	0.674	Yes*	Yes
4	North	Ameresco, Inc.	John A. Sutter Middle School	LADWP	0.510	Yes*	Yes
4	North	Ameresco, Inc.	Sherman Oaks Center for Enriched Studies (SOCES) Magnet	LADWP	0.527	Yes*	Yes
5	East	Centrica	Bell Senior High School	SCE	0.959	No	No
6	North	Ameresco, Inc.	Pacoima Middle School	LADWP	0.588	Yes*	Yes
6	North	Ameresco, Inc.	Sun Valley Magnet	LADWP	0.423	Yes*	Yes
7	South	Centrica	Charles Drew Middle School	SCE	0.691	Yes	No
7	South	Centrica	John C. Fremont High School	LADWP	1.390	No	No
7	East	Centrica	Los Angeles Academy Middle School	LADWP	0.791	Yes	No
7	South	Centrica	Thomas A Edison Middle School	SCE	0.631	Yes	No

10.612

*This site includes two Level 2 EV charging stations in addition to the EV charging infrastructure.

EXHIBIT B

RESOLUTION TO MAKE FINDINGS PURSUANT TO GOVERNMENT CODE SECTION 4217.10 ET SEQ., AND AUTHORIZE ENTERING INTO AN ENERGY SERVICES CONTRACT WITH AMERSCO, INC. AND CENTRICA BUSINESS SOLUTIONS SERVICES, INC.

WHEREAS, Ameresco, Inc. and Centrica Business Solutions Services, Inc. (“**Energy Contractors**”) have represented to the Los Angeles Unified School District (“**District**”) that Energy Contractors have developed certain procedures for the design and installation of energy conservation measures as defined in Government Code section 4217.11 (“**Energy Conservation Measures**”); and

WHEREAS, Energy Contractors have analyzed the energy needs of the following District school sites:

- **Bell Senior High School, 4328 Bell Ave., Bell, CA 90035**
- **Benjamin Franklin High School, 820 N. Avenue 54, Los Angeles, CA 90042**
- **Charles Drew Middle School, 8511 Compton Ave., Los Angeles, CA 90001**
- **Crenshaw Magnets: Science Tech Engineer Math & Medicine, 5010 11th Ave., Los Angeles, CA 90043**
- **John C. Fremont High School, 7676 South San Pedro Street, Los Angeles, CA 90003**
- **John Muir Middle School, 5929 S. Vermont Ave., Los Angeles, CA 90044**
- **Los Angeles Academy Middle School, 644 E. 56th St., Los Angeles, CA 90011**
- **Susan Miller Dorsey Senior High, 3537 Farmdale Ave., Los Angeles, CA 90016**
- **Thomas A Edison Middle School, 6500 Hooper Ave., Los Angeles, CA 90001**
- **John A. Sutter Middle School, 7330 Winnetka Ave., Winnetka, CA 91306**
- **Northridge Middle School, 17960 Chase St., Northridge, CA 91325**
- **Pacoima Middle School, 9919 Laurel Canyon Blvd., Pacoima, CA 91331**
- **Sherman Oaks Center for Enriched Studies (SOCES) Magnet, 18605 Erwin St., Reseda, CA 91335**
- **Sun Valley Magnet, 7330 Bakman Ave., Sun Valley, CA 91352**

(collectively “**School Sites**”) and have represented that Energy Contractors’ providing the Energy Conservation Measures at the School Sites will result in a reduction in energy consumption or demand that will result in net cost savings to the District (“**Cost Savings**”). A list of the School Sites and an energy analysis is attached as **Attachment A** and incorporated herein (“**School Sites List**” and “**Energy Analysis**”); and

WHEREAS, Energy Contractors have represented to the District that Energy Contractors can provide, install, and construct the Energy Conservation Measures; and

WHEREAS, based upon the Energy Analysis and presentation by District staff and its consultants, the anticipated cost of \$98.4 million to the District for Energy Contractors to design, provide, install, construct and maintain the Energy Conservation Measures will be less than the anticipated marginal cost of \$133.5 million to the District of thermal, electrical, or other energy that would

have been consumed by the District in the absence of the Energy Conservation Measures resulting in a cost savings of \$35.1 million; and

WHEREAS, Government Code section 4217.12 (a) authorizes a public agency to enter into an energy services contract if the determination is made at a regularly scheduled public hearing, public notice of which is given at least two weeks in advance, and if the governing body finds that the anticipated cost to the public agency for thermal or electrical energy or conservation services provided by the energy conservation facility under the contract will be less than the anticipated marginal cost to the public agency of thermal, electrical, or other energy that would have been consumed by the public agency in the absence of those purchases; and

WHEREAS, on this date, pursuant to Government Code section 4217.10 et seq., the District's Governing Board held a public hearing with respect to the District entering into energy services contracts with Energy Contractors ("**Energy Services Contract**"); and

WHEREAS, the Governing Board has determined that the anticipated cost of the energy consumption or demand to the District pursuant to an Energy Services Contract will be less than the anticipated marginal cost to the District in the absence of the Energy Conservation Measures and an Energy Services Contract; and

WHEREAS, the District desires to retain Energy Contractors to design, provide, install, and construct the Energy Conservation Measures pursuant to the terms and conditions of an Energy Services Contract.

NOW, THEREFORE, it is found, determined, and resolved by the Governing Board of the District as follows:

1. That the District held a public hearing at a regularly scheduled meeting of the Governing Board.
2. Based upon reports of staff, reviewed by the Governing Board in connection herewith, and pursuant to Government Code section 4217.12, the anticipated cost of the energy consumption or demand to the District pursuant to an Energy Services Contract will be less than the anticipated marginal cost to the District in the absence of the Energy Conservation Measures and an Energy Services Contract.
3. It is in the best interest of the District to enter an Energy Services Contract with the Energy Contractors.
4. That the District's superintendent or designee is authorized to enter into an Energy Service Contract with the Energy Contractors for the sites listed in the School Sites List and Energy Analysis attached as **Attachment A**, subject to the District's standard terms and conditions approved by the Office of the General Counsel.

PASSED AND ADOPTED by the Board of Education of the Los Angeles Unified School District,
on _____, 20____, by the following vote:

AYES:

NOES:

ABSTAINED:

ABSENT:

CERTIFICATION

I, _____ Executive Officer of the Board of Education of the City of Los Angeles, Los Angeles County, State of California, do hereby certify that the foregoing Resolution was duly approved and adopted by the Board of Education of the City of Los Angeles at a meeting after a hearing that was held on _____, 20____, with a copy of the Resolution being on file in the District's Office of the Executive Officer.

Executive Officer, Board of
Education of the City of Los Angeles

Attachments:

Attachment A – School Sites List and Energy Analysis

ATTACHMENT A – School Sites List and Energy Analysis

School Site:

No.	School	Address	PV System Size (MW)
1	Bell Senior High School	4328 Bell Ave., Bell, CA 90035	0.959
2	Benjamin Franklin High School	820 N. Avenue 54, Los Angeles, CA 90042	0.763
3	Charles Drew Middle School	8511 Compton Ave., Los Angeles, CA 90001	0.691
4	Crenshaw Magnets: Science Tech Engineer Math & Medicine	5010 11 th Ave., Los Angeles, CA 90043	1.207
5	John C. Fremont High School	7676 South San Pedro Street, Los Angeles, CA 90003	1.390
6	John Muir Middle School	5929 S. Vermont Ave., Los Angeles, CA 9004	0.612
7	Los Angeles Academy Middle School	644 E. 56 th St., Los Angeles, CA 90011	0.791
8	Susan Miller Dorsey Senior High	3537 Farmdale Ave., Los Angeles, CA 90016	0.846
9	Thomas A Edison Middle School	6500 Hooper Ave., Los Angeles, CA 90001	0.631
10	John A. Sutter Middle School	7330 Winnetka Ave., Winnetka, CA 91306	0.510
11	Northridge Middle School	17960 Chase St., Northridge, CA 91325	0.674
12	Pacoima Middle School	9919 Laurel Canyon Blvd., Pacoima, CA 91331	0.588
13	Sherman Oaks Center for Enriched Studies (SOCES) Magnet	18605 Erwin St., Reseda, CA 91335	0.527
14	Sun Valley Magnet	7330 Bakman Ave., Sun Valley, CA 91352	0.423

Energy Analysis:

1. The Energy Contractors have agreed to finance the Energy Conservation Measures through an Energy Services Agreement which includes approximately 15% for District project soft costs. Therefore, the anticipated up front cost to the District for an Energy Services Contract is zero. Also, the proposed costs include the Investment Tax Credits available to the District, which can be up to 40% of eligible costs.
2. The projected net cost savings to the District is anticipated to be approximately \$35.1 million over the 25-year equipment life.

Assumptions:

- It is assumed that the solar photovoltaic panels have an equipment life of at least 25 years.
- Electricity rate escalation: 3%
- PV degradation rate: 0.5%

OVERVIEW

Energy Contractor	School Site	Agreement Term Length (Years)	25 Year Est. Cumulative Electricity Cost (3% Escalation)	Total Term Payments to Energy Contractors	25 Year Net Cost Savings
Centrica	Bell Senior High School	20	\$12,002,921	\$7,784,839	\$4,218,082
Centrica	Crenshaw Magnets: Science Tech Engineer Math & Medicine	20	\$14,867,756	\$10,240,044	\$4,627,712
Centrica	Susan Miller Dorsey Senior High	20	\$9,704,221	\$7,316,965	\$2,387,256
Centrica	Charles Drew Middle School	20	\$8,047,624	\$6,171,934	\$1,875,690
Centrica	Thomas A Edison Middle School	20	\$7,612,471	\$5,563,248	\$2,049,223
Centrica	Benjamin Franklin High School	20	\$9,348,745	\$7,026,377	\$2,322,368
Centrica	John C. Fremont High School	20	\$15,790,588	\$11,229,581	\$4,561,007
Centrica	Los Angeles Academy Middle School	20	\$9,090,695	\$6,548,938	\$2,541,757
Centrica	John Muir Middle School	20	\$7,051,490	\$5,161,649	\$1,889,841
Ameresco, Inc.	Northridge Middle School	25	\$9,310,926	\$7,494,087	\$1,816,839
Ameresco, Inc.	Pacoima Middle School	25	\$8,473,546	\$6,481,452	\$1,992,094
Ameresco, Inc.	Sherman Oaks Center for Enriched Studies (SOCES) Magnet	25	\$7,914,256	\$6,229,500	\$1,684,756
Ameresco, Inc.	Sun Valley Magnet	25	\$6,722,040	\$4,886,327	\$1,835,713
Ameresco, Inc.	John A. Sutter Middle School	25	\$7,621,406	\$6,287,085	\$1,334,321
TOTAL			\$133,558,685	\$98,422,026	\$35,136,659

**Cumulative Annual Net Cost Savings
14 School Sites**

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	16,832,157	\$3,842,959	\$4,412,760	-\$569,799
2	16,757,442	\$5,124,126	\$4,644,815	\$479,312
3	16,679,843	\$4,332,658	\$4,653,757	-\$321,099
4	16,602,605	\$4,433,038	\$4,667,991	-\$234,953
5	16,525,723	\$4,535,979	\$4,679,142	-\$143,164
6	16,449,197	\$4,641,864	\$4,692,222	-\$50,358
7	16,373,027	\$4,453,862	\$4,706,005	-\$252,143
8	16,297,211	\$4,565,148	\$4,718,308	-\$153,159
9	16,221,747	\$4,679,225	\$4,735,049	-\$55,823
10	16,146,633	\$4,796,163	\$4,747,858	\$48,306
11	16,071,868	\$4,916,037	\$4,762,744	\$153,290
12	15,997,449	\$5,038,914	\$4,776,437	\$262,475
13	15,923,376	\$5,164,730	\$4,792,807	\$371,922
14	15,849,649	\$5,293,850	\$4,807,532	\$486,315
15	15,776,260	\$5,426,208	\$4,824,483	\$601,728
16	15,703,214	\$5,561,888	\$4,841,339	\$720,549
17	15,630,506	\$5,700,971	\$4,860,876	\$840,095
18	15,558,137	\$5,843,544	\$4,878,729	\$964,814
19	15,486,103	\$5,989,539	\$4,894,722	\$1,094,814
20	15,414,402	\$6,139,354	\$1,671,681	\$4,467,673
21	15,343,036	\$6,292,930	\$1,312,697	\$4,980,233
22	15,271,999	\$6,450,364	\$1,321,397	\$5,128,967
23	15,201,291	\$6,611,743	\$1,330,323	\$5,281,419
24	15,130,914	\$6,777,009	\$1,339,481	\$5,437,528
25	15,060,862	\$6,946,591	\$1,348,877	\$5,597,714
	398,304,654	\$133,558,685	\$98,422,026	\$35,136,659

Bell Senior High School

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	1,509,445	\$ 349,738	\$373,120	-\$23,382
2	1,502,652	\$ 358,610	\$402,354	-\$43,745
3	1,495,891	\$ 367,706	\$402,305	-\$34,600
4	1,489,159	\$ 377,033	\$403,946	-\$26,914
5	1,482,458	\$ 386,596	\$404,182	-\$17,586
6	1,475,787	\$ 396,402	\$405,062	-\$8,660
7	1,469,146	\$ 406,457	\$405,538	\$918
8	1,462,535	\$ 416,766	\$406,612	\$10,154
9	1,455,953	\$ 427,338	\$408,237	\$19,101
10	1,449,401	\$ 438,177	\$408,366	\$29,811
11	1,442,879	\$ 449,292	\$410,047	\$39,244
12	1,436,386	\$ 460,688	\$410,186	\$50,501
13	1,429,922	\$ 472,373	\$411,832	\$60,541
14	1,423,488	\$ 484,355	\$412,890	\$71,464
15	1,417,082	\$ 496,641	\$413,362	\$83,279
16	1,410,705	\$ 509,238	\$414,248	\$94,990
17	1,404,357	\$ 522,155	\$415,501	\$106,654
18	1,398,037	\$ 535,399	\$417,075	\$118,324
19	1,391,746	\$ 548,980	\$417,924	\$131,055
20	1,385,483	\$ 562,904	\$42,049	\$520,855
21	1,379,249	\$ 577,182	\$0	\$577,182
22	1,373,042	\$ 591,823	\$0	\$591,823
23	1,366,863	\$ 606,834	\$0	\$606,834
24	1,360,712	\$ 622,227	\$0	\$622,227
25	1,354,589	\$ 638,009	\$0	\$638,009
	35,766,969	\$12,002,921	\$7,784,839	\$4,218,082

Crenshaw Magnets: Science Tech Engineer Math & Medicine

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	1,905,910	\$ 433,213	\$486,935	-\$53,722
2	1,897,333	\$ 444,202	\$530,049	-\$85,847
3	1,888,795	\$ 455,469	\$530,631	-\$75,162
4	1,880,296	\$ 467,022	\$531,766	-\$64,744
5	1,871,834	\$ 478,868	\$533,405	-\$54,537
6	1,863,411	\$ 491,014	\$533,502	-\$42,488
7	1,855,026	\$ 503,469	\$535,106	-\$31,637
8	1,846,678	\$ 516,240	\$536,123	-\$19,883
9	1,838,368	\$ 529,334	\$537,553	-\$8,219
10	1,830,096	\$ 542,760	\$538,350	\$4,411
11	1,821,860	\$ 556,528	\$539,515	\$17,013
12	1,813,662	\$ 570,644	\$540,001	\$30,643
13	1,805,500	\$ 585,118	\$540,809	\$44,309
14	1,797,376	\$ 599,960	\$541,894	\$58,066
15	1,789,287	\$ 615,178	\$543,208	\$71,970
16	1,781,236	\$ 630,782	\$545,704	\$85,077
17	1,773,220	\$ 646,782	\$547,291	\$99,491
18	1,765,240	\$ 663,187	\$547,967	\$115,220
19	1,757,297	\$ 680,009	\$549,735	\$130,274
20	1,749,389	\$ 697,257	\$50,501	\$646,756
21	1,741,517	\$ 714,943	\$0	\$714,943
22	1,733,680	\$ 733,078	\$0	\$733,078
23	1,725,878	\$ 751,672	\$0	\$751,672
24	1,718,112	\$ 770,739	\$0	\$770,739
25	1,710,380	\$ 790,288	\$0	\$790,288
	45,161,383	\$14,867,756	\$10,240,044	\$4,627,712

Susan Miller Dorsey Senior High

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	1,288,785	\$ 282,759	\$352,170	-\$69,411
2	1,282,985	\$ 289,932	\$378,013	-\$88,081
3	1,277,212	\$ 297,286	\$378,396	-\$81,111
4	1,271,465	\$ 304,826	\$379,468	-\$74,642
5	1,265,743	\$ 312,558	\$380,181	-\$67,623
6	1,260,047	\$ 320,486	\$380,537	-\$60,051
7	1,254,377	\$ 328,615	\$381,536	-\$52,920
8	1,248,732	\$ 336,951	\$382,130	-\$45,179
9	1,243,113	\$ 345,498	\$383,322	-\$37,824
10	1,237,519	\$ 354,261	\$384,063	-\$29,802
11	1,231,950	\$ 363,247	\$385,355	-\$22,108
12	1,226,406	\$ 372,461	\$386,151	-\$13,690
13	1,220,887	\$ 381,908	\$386,452	-\$4,544
14	1,215,394	\$ 391,595	\$387,259	\$4,336
15	1,209,924	\$ 401,528	\$388,525	\$13,004
16	1,204,480	\$ 411,713	\$389,203	\$22,510
17	1,199,059	\$ 422,156	\$390,294	\$31,862
18	1,193,664	\$ 432,864	\$391,752	\$41,112
19	1,188,292	\$ 443,844	\$392,530	\$51,314
20	1,182,945	\$ 455,102	\$39,629	\$415,472
21	1,177,622	\$ 466,645	\$0	\$466,645
22	1,172,322	\$ 478,482	\$0	\$478,482
23	1,167,047	\$ 490,618	\$0	\$490,618
24	1,161,795	\$ 503,063	\$0	\$503,063
25	1,156,567	\$ 515,823	\$0	\$515,823
	30,538,332	\$9,704,221	\$7,316,965	\$2,387,256

Charles Drew Middle School

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	1,110,274	\$ 234,490	\$303,025	-\$68,535
2	1,105,278	\$ 240,438	\$318,139	-\$77,701
3	1,100,304	\$ 246,536	\$318,651	-\$72,114
4	1,095,353	\$ 252,790	\$318,897	-\$66,107
5	1,090,424	\$ 259,202	\$319,878	-\$60,676
6	1,085,517	\$ 265,776	\$320,547	-\$54,771
7	1,080,632	\$ 272,518	\$321,904	-\$49,386
8	1,075,769	\$ 279,430	\$321,903	-\$42,473
9	1,070,928	\$ 286,518	\$322,592	-\$36,074
10	1,066,109	\$ 293,786	\$322,924	-\$29,138
11	1,061,311	\$ 301,237	\$323,899	-\$22,662
12	1,056,535	\$ 308,878	\$324,471	-\$15,593
13	1,051,781	\$ 316,713	\$325,641	-\$8,928
14	1,047,048	\$ 324,746	\$326,361	-\$1,615
15	1,042,336	\$ 332,984	\$327,633	\$5,351
16	1,037,646	\$ 341,430	\$328,410	\$13,020
17	1,032,976	\$ 350,090	\$329,692	\$20,398
18	1,028,328	\$ 358,970	\$330,434	\$28,536
19	1,023,701	\$ 368,075	\$330,635	\$37,440
20	1,019,094	\$ 377,412	\$36,298	\$341,114
21	1,014,508	\$ 386,985	\$0	\$386,985
22	1,009,943	\$ 396,801	\$0	\$396,801
23	1,005,398	\$ 406,865	\$0	\$406,865
24	1,000,874	\$ 417,186	\$0	\$417,186
25	996,370	\$ 427,768	\$0	\$427,768
	26,308,435	\$8,047,624	\$6,171,934	\$1,875,690

Thomas Edison Middle School

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	1,001,854	\$ 221,810	\$276,323	-\$54,512
2	997,346	\$ 227,437	\$286,914	-\$59,477
3	992,858	\$ 233,206	\$286,872	-\$53,666
4	988,390	\$ 239,121	\$288,609	-\$49,488
5	983,942	\$ 245,186	\$288,028	-\$42,842
6	979,514	\$ 251,405	\$289,227	-\$37,822
7	975,106	\$ 257,782	\$289,111	-\$31,328
8	970,718	\$ 264,321	\$290,726	-\$26,406
9	966,350	\$ 271,025	\$289,980	-\$18,955
10	962,002	\$ 277,900	\$291,967	-\$14,067
11	957,673	\$ 284,949	\$291,546	-\$6,597
12	953,363	\$ 292,177	\$292,812	-\$635
13	949,073	\$ 299,588	\$293,670	\$5,917
14	944,802	\$ 307,187	\$294,122	\$13,065
15	940,551	\$ 314,978	\$295,168	\$19,811
16	936,318	\$ 322,968	\$295,761	\$27,207
17	932,105	\$ 331,160	\$296,901	\$34,259
18	927,910	\$ 339,560	\$297,543	\$42,017
19	923,735	\$ 348,173	\$298,686	\$49,487
20	919,578	\$ 357,004	\$29,284	\$327,721
21	915,440	\$ 366,060	\$0	\$366,060
22	911,320	\$ 375,345	\$0	\$375,345
23	907,219	\$ 384,865	\$0	\$384,865
24	903,137	\$ 394,627	\$0	\$394,627
25	899,073	\$ 404,637	\$0	\$404,637
	23,739,375	\$7,612,471	\$5,563,248	\$2,049,223

Benjamin Franklin High School

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	1,230,960	\$ 272,535	\$338,558	-\$66,023
2	1,224,805	\$ 279,307	\$361,790	-\$82,482
3	1,219,294	\$ 286,392	\$362,624	-\$76,232
4	1,213,807	\$ 293,656	\$363,149	-\$69,493
5	1,208,345	\$ 301,105	\$363,365	-\$62,261
6	1,202,907	\$ 308,742	\$364,273	-\$55,531
7	1,197,494	\$ 316,573	\$365,827	-\$49,254
8	1,192,105	\$ 324,603	\$365,979	-\$41,376
9	1,186,741	\$ 332,837	\$367,778	-\$34,941
10	1,181,400	\$ 341,279	\$368,129	-\$26,850
11	1,176,084	\$ 349,936	\$369,081	-\$19,145
12	1,170,792	\$ 358,812	\$370,587	-\$11,776
13	1,165,523	\$ 367,913	\$371,601	-\$3,688
14	1,160,278	\$ 377,245	\$372,124	\$5,121
15	1,155,057	\$ 386,814	\$373,156	\$13,658
16	1,149,859	\$ 396,626	\$374,651	\$21,975
17	1,144,685	\$ 406,686	\$375,562	\$31,124
18	1,139,534	\$ 417,002	\$376,891	\$40,111
19	1,134,406	\$ 427,579	\$377,591	\$49,988
20	1,129,301	\$ 438,424	\$43,662	\$394,762
21	1,124,219	\$ 449,545	\$0	\$449,545
22	1,119,160	\$ 460,948	\$0	\$460,948
23	1,114,124	\$ 472,640	\$0	\$472,640
24	1,109,111	\$ 484,628	\$0	\$484,628
25	1,104,120	\$ 496,921	\$0	\$496,921
	29,154,112	\$9,348,745	\$7,026,377	\$2,322,368

John C. Fremont High School

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	2,182,650	\$ 460,103	\$532,818	-\$72,715
2	2,172,828	\$ 471,773	\$583,292	-\$111,518
3	2,163,050	\$ 483,740	\$583,672	-\$99,932
4	2,153,317	\$ 496,010	\$584,555	-\$88,545
5	2,143,627	\$ 508,591	\$585,894	-\$77,303
6	2,133,980	\$ 521,491	\$587,641	-\$66,150
7	2,124,377	\$ 534,719	\$587,751	-\$53,032
8	2,114,818	\$ 548,282	\$588,271	-\$39,988
9	2,105,301	\$ 562,189	\$590,154	-\$27,965
10	2,095,827	\$ 576,449	\$590,308	-\$13,859
11	2,086,396	\$ 591,071	\$591,779	-\$709
12	2,077,007	\$ 606,063	\$592,475	\$13,588
13	2,067,661	\$ 621,436	\$594,395	\$27,041
14	2,058,356	\$ 637,199	\$595,447	\$41,752
15	2,049,094	\$ 653,362	\$596,630	\$56,732
16	2,039,873	\$ 669,934	\$596,899	\$73,036
17	2,030,693	\$ 686,927	\$599,254	\$87,673
18	2,021,555	\$ 704,351	\$600,554	\$103,797
19	2,012,458	\$ 722,217	\$600,800	\$121,416
20	2,003,402	\$ 740,536	\$46,994	\$693,542
21	1,994,387	\$ 759,319	\$0	\$759,319
22	1,985,412	\$ 778,580	\$0	\$778,580
23	1,976,478	\$ 798,328	\$0	\$798,328
24	1,967,584	\$ 818,578	\$0	\$818,578
25	1,958,729	\$ 839,341	\$0	\$839,341
	51,718,860	\$15,790,588	\$11,229,581	\$4,561,007

Los Angeles Academy Middle School

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	1,234,884	\$ 264,883	\$317,268	-\$52,385
2	1,229,327	\$ 271,601	\$337,050	-\$65,449
3	1,223,795	\$ 278,491	\$337,307	-\$58,816
4	1,218,288	\$ 285,554	\$338,299	-\$52,745
5	1,212,806	\$ 292,798	\$338,982	-\$46,184
6	1,207,348	\$ 300,224	\$339,354	-\$39,130
7	1,201,915	\$ 307,840	\$340,418	-\$32,579
8	1,196,506	\$ 315,648	\$341,126	-\$25,478
9	1,191,122	\$ 323,654	\$342,479	-\$18,824
10	1,185,762	\$ 331,864	\$343,430	-\$11,566
11	1,180,426	\$ 340,282	\$343,979	-\$3,698
12	1,175,114	\$ 348,913	\$345,129	\$3,784
13	1,169,826	\$ 357,763	\$345,832	\$11,931
14	1,164,562	\$ 366,838	\$347,088	\$19,749
15	1,159,321	\$ 376,142	\$347,852	\$28,291
16	1,154,104	\$ 385,683	\$349,124	\$36,560
17	1,148,911	\$ 395,466	\$349,857	\$45,609
18	1,143,741	\$ 405,497	\$351,053	\$54,444
19	1,138,594	\$ 415,783	\$352,665	\$63,118
20	1,133,470	\$ 426,329	\$40,646	\$385,683
21	1,128,370	\$ 437,143	\$0	\$437,143
22	1,123,292	\$ 448,231	\$0	\$448,231
23	1,118,237	\$ 459,600	\$0	\$459,600
24	1,113,205	\$ 471,258	\$0	\$471,258
25	1,108,196	\$ 483,211	\$0	\$483,211
	29,261,124	\$9,090,695	\$6,548,938	\$2,541,757

John Muir Middle School

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	962,230	\$ 204,666	\$254,713	-\$50,046
2	961,749	\$ 210,701	\$264,120	-\$53,419
3	957,421	\$ 216,045	\$264,809	-\$48,763
4	953,113	\$ 221,525	\$265,281	-\$43,755
5	948,824	\$ 227,144	\$265,536	-\$38,392
6	944,554	\$ 232,906	\$266,576	-\$33,670
7	940,303	\$ 238,813	\$267,352	-\$28,539
8	936,072	\$ 244,871	\$267,867	-\$22,996
9	931,860	\$ 251,082	\$269,120	-\$18,038
10	927,666	\$ 257,451	\$270,065	-\$12,614
11	923,492	\$ 263,981	\$270,702	-\$6,721
12	919,336	\$ 270,677	\$271,032	-\$355
13	915,199	\$ 277,543	\$272,057	\$5,486
14	911,081	\$ 284,583	\$272,728	\$11,854
15	906,981	\$ 291,801	\$274,048	\$17,753
16	902,899	\$ 299,203	\$274,969	\$24,233
17	898,836	\$ 306,792	\$276,493	\$30,299
18	894,792	\$ 314,574	\$277,572	\$37,001
19	890,765	\$ 322,553	\$278,208	\$44,344
20	886,757	\$ 330,734	\$38,402	\$292,332
21	882,766	\$ 339,123	\$0	\$339,123
22	878,794	\$ 347,725	\$0	\$347,725
23	874,839	\$ 356,545	\$0	\$356,545
24	870,902	\$ 365,589	\$0	\$365,589
25	866,983	\$ 374,862	\$0	\$374,862
	22,888,215	\$7,051,490	\$5,161,649	\$1,889,841

Northridge Middle School

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	1,101,465	\$261,864	\$281,753	-\$19,888.26
2	1,095,958	\$505,099	\$282,972	\$222,126.67
3	1,090,478	\$333,612	\$284,223	\$49,389
4	1,085,025	\$340,160	\$285,506	\$54,653
5	1,079,600	\$346,875	\$286,823	\$60,053
6	1,074,202	\$353,826	\$288,173	\$65,653
7	1,068,831	\$301,637	\$289,558	\$12,079
8	1,063,487	\$308,935	\$290,979	\$17,956
9	1,058,170	\$316,411	\$292,437	\$23,974
10	1,052,879	\$324,070	\$293,933	\$30,138
11	1,047,615	\$331,917	\$295,468	\$36,449
12	1,042,376	\$339,955	\$297,042	\$42,912
13	1,037,165	\$348,161	\$298,658	\$49,503
14	1,031,979	\$356,597	\$300,316	\$56,281
15	1,026,819	\$365,238	\$302,018	\$63,221
16	1,021,685	\$374,091	\$303,764	\$70,328
17	1,016,576	\$383,161	\$305,556	\$77,605
18	1,011,493	\$392,453	\$307,395	\$85,057
19	1,006,436	\$401,941	\$309,283	\$92,658
20	1,001,404	\$411,692	\$311,220	\$100,472
21	996,397	\$421,683	\$313,209	\$108,473
22	991,415	\$431,917	\$315,251	\$116,666
23	986,458	\$442,402	\$317,346	\$125,056
24	981,525	\$453,111	\$319,498	\$133,614
25	976,618	\$464,116	\$321,706	\$142,410
	25,946,056	\$9,310,926	\$7,494,087	\$1,816,839

Pacoima Middle School

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	946,743	\$237,452	\$242,938	-\$5,486
2	942,009	\$480,056	\$244,047	\$236,009
3	937,299	\$307,931	\$245,184	\$62,747
4	932,613	\$313,825	\$246,350	\$67,476
5	927,950	\$319,872	\$247,545	\$72,327
6	923,310	\$326,140	\$248,771	\$77,369
7	918,693	\$273,251	\$250,028	\$23,223
8	914,100	\$279,832	\$251,317	\$28,516
9	909,529	\$286,574	\$252,638	\$33,936
10	904,982	\$293,480	\$253,994	\$39,486
11	900,457	\$300,554	\$255,384	\$45,170
12	895,955	\$307,801	\$256,810	\$50,991
13	891,475	\$315,195	\$258,273	\$56,922
14	887,017	\$322,799	\$259,773	\$63,026
15	882,582	\$330,588	\$261,312	\$69,276
16	878,169	\$338,568	\$262,891	\$75,677
17	873,779	\$346,742	\$264,511	\$82,231
18	869,410	\$355,115	\$266,172	\$88,943
19	865,063	\$363,661	\$267,877	\$95,784
20	860,737	\$372,448	\$269,626	\$102,821
21	856,434	\$381,449	\$271,421	\$110,028
22	852,151	\$390,670	\$273,263	\$117,407
23	847,891	\$400,116	\$275,153	\$124,963
24	843,651	\$409,758	\$277,092	\$132,666
25	839,433	\$419,671	\$279,082	\$140,589
	22,301,432	\$8,473,546	\$6,481,452	\$1,992,094

Sherman Oaks Center for Enriched Studies (SOCES) Magnet

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	843,661	\$221,118	\$234,061	-\$12,943
2	839,443	\$463,305	\$235,091	\$228,214
3	835,245	\$290,756	\$236,147	\$54,609
4	831,069	\$296,216	\$237,229	\$58,986
5	826,914	\$301,817	\$238,339	\$63,479
6	822,779	\$307,632	\$239,476	\$68,156
7	818,665	\$254,277	\$240,642	\$13,635
8	814,572	\$260,382	\$241,837	\$18,544
9	810,499	\$266,635	\$243,062	\$23,572
10	806,447	\$273,040	\$244,318	\$28,721
11	802,415	\$279,600	\$245,606	\$33,994
12	798,402	\$286,321	\$246,926	\$39,394
13	794,410	\$293,175	\$248,280	\$44,894
14	790,438	\$300,226	\$249,669	\$50,557
15	786,486	\$307,448	\$251,092	\$56,356
16	782,554	\$314,846	\$252,552	\$62,294
17	778,641	\$322,425	\$254,049	\$68,376
18	774,748	\$330,188	\$255,585	\$74,603
19	770,874	\$338,107	\$257,160	\$80,947
20	767,020	\$346,253	\$258,775	\$87,477
21	763,185	\$354,596	\$260,432	\$94,164
22	759,369	\$363,144	\$262,132	\$101,012
23	755,572	\$371,899	\$263,875	\$108,024
24	751,794	\$380,833	\$265,664	\$115,168
25	748,035	\$390,020	\$267,499	\$122,521
	19,873,238	\$7,914,256	\$6,229,500	\$1,684,756

Sun Valley Magnet

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	696,500	\$186,022	\$182,381	\$3,641
2	693,018	\$427,357	\$183,277	\$244,079
3	689,552	\$253,928	\$184,195	\$69,732
4	686,105	\$258,485	\$185,136	\$73,350
5	682,674	\$263,162	\$186,099	\$77,063
6	679,261	\$268,025	\$187,085	\$80,939
7	675,864	\$213,696	\$188,095	\$25,600
8	672,485	\$218,801	\$189,131	\$29,671
9	669,123	\$224,030	\$190,191	\$33,839
10	665,777	\$229,386	\$191,277	\$38,109
11	662,448	\$234,872	\$192,390	\$42,482
12	659,136	\$240,491	\$193,531	\$46,960
13	655,840	\$246,217	\$194,699	\$51,518
14	652,561	\$252,111	\$195,896	\$56,215
15	649,298	\$258,148	\$197,123	\$61,025
16	646,052	\$264,332	\$198,380	\$65,952
17	642,822	\$270,666	\$199,669	\$70,997
18	639,607	\$277,153	\$200,989	\$76,164
19	636,409	\$283,767	\$202,343	\$81,424
20	633,227	\$290,573	\$203,730	\$86,843
21	630,061	\$297,545	\$205,152	\$92,392
22	626,911	\$304,685	\$206,610	\$98,075
23	623,776	\$312,000	\$208,104	\$103,896
24	620,657	\$319,458	\$209,636	\$109,822
25	617,554	\$327,132	\$211,207	\$115,925
	16,406,720	\$6,722,040	\$4,886,327	\$1,835,713

John A. Sutter Middle School

Year	Annual Energy Production (kWh)	Annual Energy Cost Savings (\$)	Annual Payments (\$)	Net Annual Cost Savings After Payment (\$)
1	816,796	\$212,306	\$236,699	-\$24,393
2	812,712	\$454,309	\$237,707	\$216,602
3	808,648	\$281,560	\$238,741	\$42,820
4	804,605	\$286,814	\$239,800	\$47,014
5	800,582	\$292,204	\$240,885	\$51,319
6	796,579	\$297,796	\$241,998	\$55,798
7	792,596	\$244,215	\$243,138	\$1,077
8	788,633	\$250,086	\$244,307	\$5,779
9	784,690	\$256,100	\$245,505	\$10,595
10	780,767	\$262,260	\$246,734	\$15,526
11	776,863	\$268,570	\$247,993	\$20,577
12	772,979	\$275,034	\$249,284	\$25,750
13	769,114	\$281,628	\$250,607	\$31,020
14	765,268	\$288,410	\$251,965	\$36,445
15	761,442	\$295,357	\$253,356	\$42,001
16	757,635	\$302,473	\$254,783	\$47,691
17	753,846	\$309,763	\$256,246	\$53,517
18	750,077	\$317,231	\$257,746	\$59,485
19	746,327	\$324,850	\$259,285	\$65,565
20	742,595	\$332,686	\$260,863	\$71,823
21	738,882	\$340,713	\$262,482	\$78,231
22	735,188	\$348,935	\$264,142	\$84,793
23	731,512	\$357,358	\$265,845	\$91,513
24	727,854	\$365,954	\$267,591	\$98,363
25	724,215	\$374,793	\$269,383	\$105,409
	19,240,407	\$7,621,406	\$6,287,085	\$1,334,321